Two independent finite-difference solvers for an open Helmholtz resonator: verification and comparison with published measurements

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Abstract

The natural frequency of a Helmholtz resonator depends on an end correction that lumped models assume. I compute it instead with two independent axisymmetric finite-difference solvers for an open-neck resonator: a frequency-domain solver that closes the mouth with the radiation impedance of a baffled piston, and a time-domain leapfrog solver that meshes the exterior half-space and lets the resonator select its own frequency after a broadband pulse. Manufactured solutions give observed orders of 2.03 and 1.98; the time-domain test also exposed a half-cell offset of the rigid walls, an $O(h)$ geometric error that has to be extrapolated away. With Richardson extrapolation the two solvers give 203.4 and 204.6 Hz (grid convergence index 2.0% and 1.5%), a 0.6% gap. A study of three outer boundaries shows that the original sponge layer reflected 44% of the incident wave, which explained a factor-six spread in the computed radiation Q. Compared with published measurements, the solver reproduces two resonators of Selamet et al. to 0.9% and 2.2% with no fitted parameter, and misses an Erlenmeyer flask by 10–13%. The solvers are lossless, so damping is not predicted. A training-free test also shows that neural bases fit the field to better than 0.4% yet leave PDE residuals thousands of times larger than the zero field.

1 Introduction

A Helmholtz resonator is a cavity of volume V connected to the outside through a neck of length L and cross-section $S = \pi a^2$. The air plug in the neck oscillates against the compressibility of the air in the cavity, and the lossless lumped model gives

$$f_0 = \frac{c}{2\pi}\sqrt{\frac{S}{V\,(L + \Delta L)}}, \tag{1}$$

where ΔL, the end correction, accounts for the air that moves with the plug just outside each end of the neck [7, 8, 21]. ΔL is of the order of the neck radius and is usually taken from tables. For a neck as short as the one studied here ($L = 4a$), it lowers f_0 by about 15%.

A numerical model can compute ΔL instead of assuming it, but only if its own error is known. The question addressed here is therefore narrow: can a finite-difference model of an open resonator predict f_0 with a stated numerical uncertainty, and where does it stop agreeing with measurements? The contributions are:

*This work started as the author's individual TIPE (the research project of the French *classes préparatoires*) in 2023 and was rebuilt and extended in 2026. Code, data and the scripts that produce every figure and number: `https://github.com/clementdumeril/helmholtz-resonator-solver`.

- two independent axisymmetric finite-difference solvers, one in the frequency domain with an impedance condition at the mouth and one in the time domain with a meshed exterior and three interchangeable outer boundaries (sponge, first-order Mur, PML);
- a verification sequence (manufactured solutions, Richardson extrapolation with a grid convergence index, cross-solver comparison, outer-boundary study) in which the manufactured solution revealed a half-cell offset of the rigid walls;
- a comparison with three published measured resonators, two reproduced and one missed, and a training-free measurement of how well neural bases represent the verified field and its PDE residual.

2 Related work

End corrections and measured resonators. Rayleigh derived the end correction of a flanged pipe [8]; Ingard studied the theory and design of resonators, including the end corrections on both sides of the neck [7]; Morse and Ingard give the radiation impedance of a baffled piston used here [10]. Selamet et al. studied concentric cylindrical resonators with two-dimensional analytical models, a three-dimensional boundary-element method and experiments [19]; two of their configurations are used in Section 6.2. Moloney measured quality factors of glass bottles and found them slightly lower than his predictions [9]. Indenbom and Pogossian measured the resonance of a filled Erlenmeyer flask [6]; its empty case is used here.

Verification. The distinction between code verification, solution verification and validation follows Roache [17] and Oberkampf and Roy [13]. Code verification uses the method of manufactured solutions [18]. Discretisation uncertainty is reported with the grid convergence index (GCI) [2, 16].

Open boundaries. The first-order absorbing condition of Mur [12] is exact only at normal incidence. The perfectly matched layer (PML) of Bérenger [1] is reflectionless in the continuous problem for a planar interface; on curved boundaries it requires a curvilinear stretch [3]. The time-domain PML used here is the unsplit second-order form for the scalar wave equation with two auxiliary fields [4].

Neural PDE solvers. Physics-informed neural networks minimise the PDE residual over the parameters of a network [15]. Their difficulties on oscillatory problems are usually attributed to spectral bias [14, 22], with remedies such as sine activations [20] and domain decomposition [11]. Trefftz methods use functions that satisfy the homogeneous Helmholtz equation exactly [5]. Section 6.3 tests the spectral-bias explanation on this problem without training a network.

3 Problem formulation

Geometry. The resonator is axisymmetric (Table 1): a cylindrical neck of radius $a = 10$ mm and length $L = 40$ mm opens at $z = 0$ into a rigid baffle and joins a cylindrical cavity of radius 40 mm and height 80 mm. All walls are rigid. The air is lossless, with $c = 343$ m/s and $\rho = 1.204$ kg/m^3.

Equations. In the time domain the acoustic pressure obeys

$$\partial_t^2 p = c^2 \nabla^2 p + F(\mathbf{x}, t), \qquad \nabla^2 p = \partial_r^2 p + \frac{1}{r}\partial_r p + \partial_z^2 p, \tag{2}$$

Table 1: Reference geometry and numerical parameters.

	quantity	value
geometry	neck radius a / length L	10 / 40 mm
	cavity radius / height	40 / 80 mm
air	c, ρ	343 m/s, 1.204 kg/m^3
frequency domain	volume source (Gaussian, cavity centre)	width 10 mm
	peak search	0.05 Hz steps, 3-point parabola
time domain	exterior (radius × depth)	120 × 100 mm
	absorbing layer thickness	30 mm
	source	Ricker, 250 Hz centre, at $z = -55$ mm
	time step	$\Delta t = 0.5\,h/(c\sqrt{2})$
	simulated time; fit window	250 ms; $t \geq 45$ ms
grids	h	2, 1, 0.5 (0.25) mm

and in the frequency domain, for a time dependence at angular frequency ω and $k = \omega/c$,

$$\nabla^2 p + k^2 p = -F(\mathbf{x}). \tag{3}$$

On the axis, $(1/r)\,\partial_r p$ is indeterminate; since $\partial_r p = 0$ there, L'Hôpital's rule gives $(1/r)\,\partial_r p \to \partial_r^2 p$, so $\nabla^2 p = 2\,\partial_r^2 p + \partial_z^2 p$ at $r = 0$. Rigid walls impose $\partial_n p = 0$.

Radiation. The two solvers model the exterior differently, which is what makes their comparison informative. The frequency-domain solver stops at the mouth and applies the low-frequency impedance of a baffled piston [10],

$$Z_r = \rho c \left[\tfrac{1}{2}(ka)^2 + \mathrm{i}\,\tfrac{8}{3\pi} ka\right], \tag{4}$$

as a Robin condition $\partial_z p = \alpha p$ on $z = 0$, $r \leq a$, with $\alpha = \mathrm{i}\omega\rho/Z_r$. Its reactance contains the exterior end correction $8a/(3\pi)$, so this solver imposes the exterior part of ΔL. The time-domain solver meshes a baffled exterior half-space and computes the whole end correction; its outer boundary has to absorb the outgoing wave.

Reference value. For comparison, Eq. (1) is evaluated with $\Delta L = 0.85a + 0.66a = 1.51a$: the flanged exterior correction [8] plus an interior correction of $0.66a$, close to the $0.67a$ fitted with the Dirichlet mouth in Section 5.2. This gives $f_0 = 205.6$ Hz. It is a semi-empirical reference, not an independent measurement.

4 Numerical method

Both solvers use a uniform grid of step h on the meridian plane (Fig. 1) and the standard second-order stencil of the axisymmetric Laplacian, with the radial part written as $4(p_1 - p_0)/h^2$ on the axis.

Frequency domain (`src/harmonic_solver.py`). The operator of Eq. (3) is assembled as a sparse complex matrix. A rigid wall is imposed by folding the ghost node onto the diagonal: the coefficient of the missing neighbour is added to the diagonal instead of appearing off-diagonal, which enforces a zero flux across the face half a cell beyond the last fluid node. The Robin condition at the mouth eliminates the ghost node through $p_{-1} = p_1 - 2h\alpha p_0$. Because k^2 and α depend on ω, each frequency requires a new sparse LU factorisation (SciPy `spsolve`): 0.1 s on the 1 mm grid (4 961 nodes), 4 s on the 0.25 mm grid (77 441 nodes). A Gaussian volume source of width 10 mm drives the cavity; f_0 is the peak of the cavity pressure, located by a

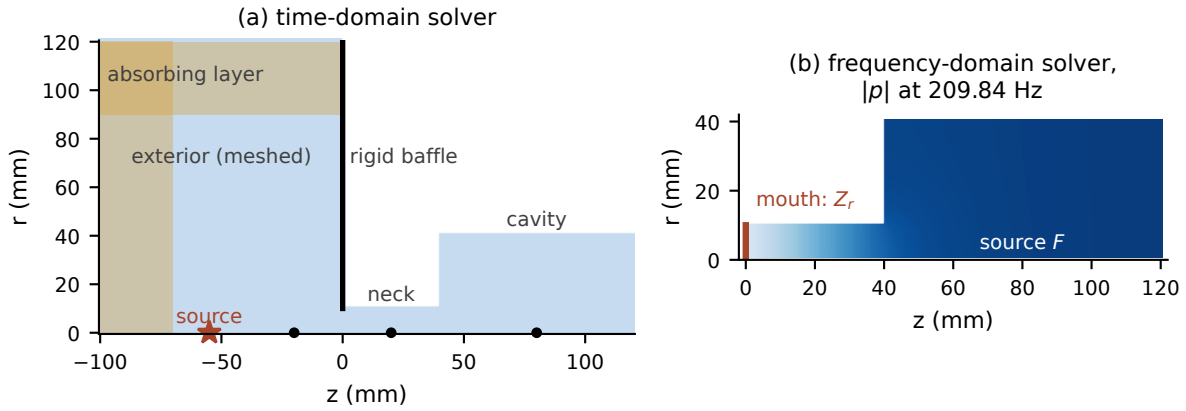


Figure 1: Computational domains on the meridian plane. (a) Time-domain solver: open neck, meshed exterior with a rigid baffle at $z = 0$, absorbing layer (shaded) on the outer boundaries, Ricker source (star) and the three probes (dots). (b) Frequency-domain solver: the domain stops at the mouth, which carries the radiation impedance; colour shows $|p|$ at 209.84 Hz on the 0.5 mm grid (`data/helmholtz_mode.npz`).

three-point parabola on a 0.05 Hz sweep, and the radiation Q is f_0 over the -3 dB width. A second configuration of the same scheme, with the mouth set to $p = 1$ (Dirichlet), is used in `notebooks/harmonic_study.ipynb` for the neck-length study and for Selamet's side-branch resonators, whose mouth is a pressure node at resonance.

Time domain (`src/transient_solver.py`). Equation (2) is advanced by explicit leapfrog with $\Delta t = 0.5\,h/(c\sqrt{2})$. The Laplacian is written in masked finite volumes: fluxes through faces between a fluid and a solid cell are zero, which again places each wall on the face half a cell beyond the last node. The neck opens onto an exterior of radius 120 mm and depth 100 mm terminated by 30 mm absorbing layers. A Ricker pulse centred at 250 Hz is emitted 55 mm in front of the mouth, so the resonator is excited over a broad band and rings at its own frequency after the pulse has passed. f_0 and Q are obtained by fitting $Ae^{-t/\theta}\cos(2\pi f_0 t + \phi)$ to the cavity probe for $t \geq 45$ ms (`verification/grid_convergence.py`), with $Q = \pi f_0 \theta$. The fit locates f_0 well below the 4.9 Hz resolution of an FFT over the same window.

Outer boundaries (`studies/radiation_boundaries.py`). Three terminations share one solver core: (i) a sponge, i.e. a damping term $\sigma\,\partial_t p$ with $\sigma = \sigma_{\max} d^2$ across the layer, as in the original solver; (ii) the first-order Mur condition $\partial_t p + c\,\partial_n p = 0$ on the outer ring [12]; (iii) the unsplit PML [4]

$$\begin{aligned} \partial_t^2 p + (s_r + s_z)\,\partial_t p + s_r s_z\, p &= c^2 \nabla^2 p + \nabla \cdot \boldsymbol{\psi}, \\ \partial_t \psi_r + s_r \psi_r = c^2 (s_z - s_r)\,\partial_r p, \qquad & \partial_t \psi_z + s_z \psi_z = c^2 (s_r - s_z)\,\partial_z p, \end{aligned} \tag{5}$$

with quadratic profiles s_r, s_z. This is the Cartesian stretch applied in the (r, z) plane: the $1/r$ term of a cylindrical PML [3] is left out.

Neural-basis test (`studies/neural_basis_benchmark.py`). For each family of 96 functions ϕ_j at random initialisation (tanh MLP, SIREN with $\omega_0 = 5$ and 8, Gabor atoms, a three-zone partition of unity of sine networks in the spirit of FBPINNs, and a Trefftz basis $J_0(\kappa_m r) Z_m(z)$), the frequency-domain field P at 209.84 Hz on the 1 mm grid is projected onto the span of $\{\phi_j\}$ by least squares. Two numbers are reported: the relative field error of that projection, and the residual $\|\nabla^2 \tilde{p} + k^2 \tilde{p} + F\|/\|F\|$ of the projection $\tilde{p}$ computed with automatic differentiation on 900 points of the fluid domain. A residual of 100% is what the zero field achieves. No network is trained in this test: it measures the best that a basis allows, not what an optimiser finds.

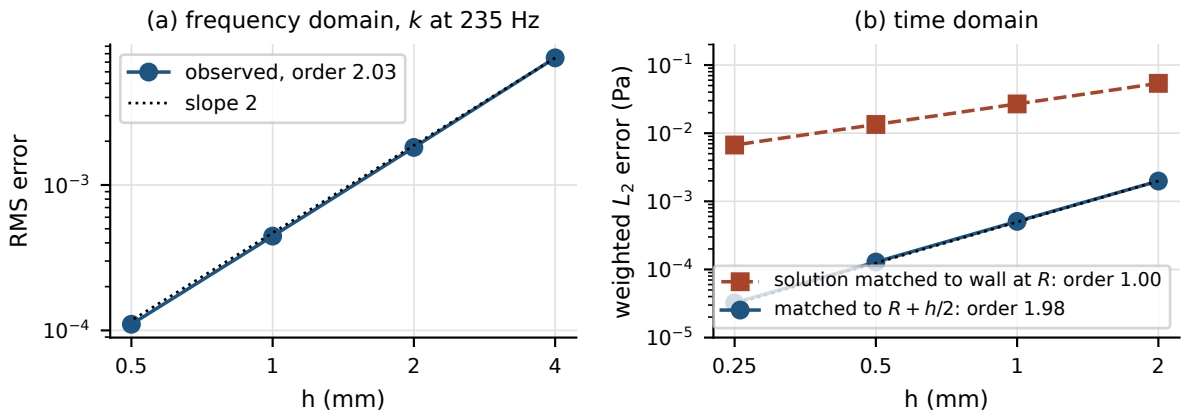


Figure 2: Manufactured solutions. (a) Frequency-domain scheme on a smooth domain with Dirichlet edges (`data/mms_results.npz`). (b) Time-domain scheme with rigid walls (`data/mms_transient.npz`): matched to the nominal wall position the error converges at first order; matched to the wall at $R + h/2$ it converges at second order.

5 Verification

5.1 Code verification: manufactured solutions

For the frequency-domain scheme, the solution $p_e = \cos(\pi r/R)\cos(\pi z/H)$ on a 40×120 mm rectangle, regular on the axis, is imposed through its source term at the wavenumber of 235 Hz. Over four grids from 4 to 0.5 mm the RMS error decreases at order 2.03 (Fig. 2a). This test uses Dirichlet edges and therefore does not exercise the wall treatment.

The time-domain test does. It uses $p = J_0(\lambda r)\cos\big(k_z(z + z_0)\big)\,T(t)$ in a closed cylinder, with λ chosen so that $\partial_r p = 0$ on the lateral wall and k_z so that $\partial_z p = 0$ on both ends, and $T(t) = u^3 e^{-u}$, $u = t/\tau$, with τ matched to the wave period so that $\partial_t^2 p$ and $c^2\nabla^2 p$ are of the same size. When the manufactured solution places the walls at the nominal radius R and height H, the observed order is 1.00 on four grids (Fig. 2b). When it places them at $R + h/2$ and $H + h$, the observed order is 1.98 and the error at $h = 2$ mm is 27 times smaller. The scheme is second order; the first-order error comes from the wall position. The frequency-domain scheme has the same convention through its ghost-node folding. Both solvers therefore simulate a resonator whose dimensions are enlarged by $O(h)$, and raw frequencies at a single grid carry a bias of a few percent.

5.2 Solution verification: grid convergence

On three grids with refinement ratio 2, the observed order p, the extrapolated value and the fine-grid GCI are [2, 16]

$$p = \frac{\ln\left|(f_3 - f_2)/(f_2 - f_1)\right|}{\ln 2}, \quad f_{h\to 0} = f_1 + \frac{f_1 - f_2}{2^p - 1}, \quad \mathrm{GCI} = 1.25\,\frac{|f_2 - f_1|/f_1}{2^p - 1}, \tag{6}$$

with f_1 on the finest grid. Table 2 and Fig. 3 give the result on the reference geometry. Both solvers converge at an observed order close to 1, as expected from the wall offset and from the staircase representation of the neck–cavity junction. Extrapolated, they give 203.39 and 204.60 Hz, 0.59% apart, which is inside both uncertainty bands. Their raw 1 mm values already agree to 0.3%, but both carry the same geometric bias, so that agreement alone says little; only the extrapolated values are compared.

Fourth grid. Three grids give exactly one observed order, so they cannot show that the solution is in the asymptotic range. For the frequency-domain solver I added a 0.25 mm

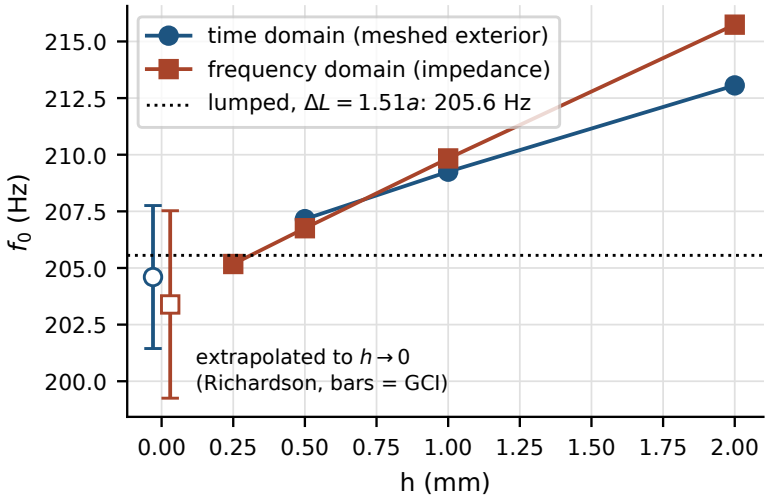


Figure 3: Natural frequency of the reference resonator against grid step for both solvers (`data/convergence_f0.npz`, `data/fdm_sweep_h*.npz`), with the Richardson extrapolations at $h = 0$ and their GCI, and the semi-empirical lumped reference.

Table 2: Grid convergence of f_0 (Hz) on the reference geometry. Time domain: damped-sinusoid fit; frequency domain: peak of the response.

solver	$h = 2$	$h = 1$	$h = 0.5$	$h = 0.25$ mm	p	$f_{h\to 0}$	GCI
time domain (meshed exterior)	213.06	209.25	207.16	–	0.86	204.60	1.5%
frequency domain (impedance)	215.74	209.84	206.75	205.18	0.94	203.39	2.0%
triplet 1 / 0.5 / 0.25 mm					0.96	203.52	1.01%

grid (`verification/control_grid.py`). The second triplet gives $p = 0.96$ and moves the extrapolated value by 0.06%, well inside the GCI. The same check was done in the harmonic notebook with the Dirichlet mouth for $L = 2$ and 4 cm: the observed order goes from 0.91 to 1.00 and from 0.87 to 1.01, and the extrapolations move by 0.20% and 0.36%. A 0.25 mm time-domain run would take about 8 h on one core and was not done; the time-domain extrapolation therefore rests on three grids.

Neck length. With the Dirichlet mouth, the same three-grid study over five neck lengths from 2 to 6 cm gives observed orders between 0.87 and 0.95 and GCIs between 2.0% and 2.3% (mean 2.1%; `notebooks/harmonic_study.ipynb`, section 4). Fitted on those extrapolated frequencies, a free power law AL^b and the effective-length model $A/\sqrt{L + \Delta L}$ cannot be told apart (ΔAICc $= 0.3$); the effective-length fit gives $\Delta L = 0.67a$, the interior correction used in the reference value.

5.3 Outer boundary of the time-domain solver

The frequency was insensitive to the outer boundary, but the damping was not: with the original sponge layer, the radiation Q ranged from 44 to 267 as the layer strength, thickness and domain size were changed on the 2 mm grid, while f_0 did not move (`data/sponge_sensitivity.npz`). To find out why, each termination was tested on an empty exterior box: a pulse is fired towards the boundary and the probe signal is compared with the same run on a domain four times larger, from which nothing can return within the 1.4 ms window. The difference is the spurious reflection, with no analytic reference needed.

The sponge reflected 44% of the incident amplitude (Table 3, Fig. 4a), enough to explain the

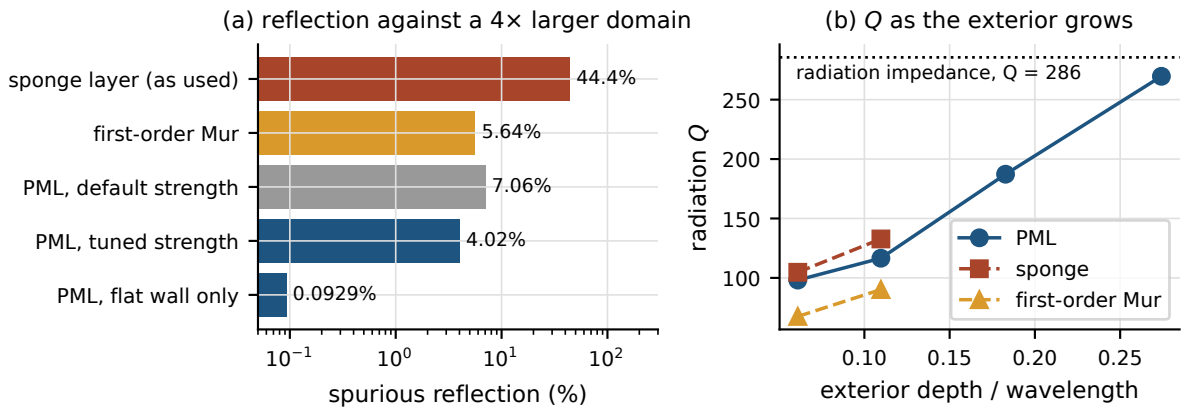


Figure 4: Outer boundaries (`data/pml_study.npz`). (a) Spurious reflection of a 4 kHz-centred Ricker pulse, measured at an off-axis probe against the same run on a domain four times larger. (b) Radiation Q of the resonator as the exterior grows, against the Q of the frequency-domain solver with the analytic impedance.

Table 3: Outer-boundary treatments. Reflection: peak spurious signal relative to the peak incident signal. Drift: change in the radiation Q of the resonator (1 mm grid) when the exterior depth goes from 10 to 18 cm.

treatment	spurious reflection	Q drift, 10 → 18 cm
sponge layer (original solver)	44%	+26.8%
first-order Mur	5.6%	+33.5%
PML (tuned strength)	4.0%	−1.8%

spread in Q. The PML reflects 7.1% at the default strength and 4.0% after scanning it, and its Q moves by −1.8% between the two exterior sizes, against +26.8% for the sponge and +33.5% for the Mur condition. Running each wall alone isolates the limit of the PML: 0.093% (−61 dB) on the flat boundary $z = \text{const}$, where the Cartesian stretch is exact, and 4.0% (−28 dB) on the cylindrical boundary, where the $1/r$ term is missing. Neither a stronger profile nor a thicker layer lowers that floor.

A stable Q between 10 and 18 cm is still not convergence. At 209 Hz the wavelength is 1.64 m, so the 10 cm exterior is 0.061 of a wavelength and the layer sits in the reactive near field of the mouth. Growing the exterior to 0.110, 0.183 and 0.274 wavelengths takes Q from 98.1 to 116.6, 187.2 and 269.6 (Fig. 4b), approaching the $Q = 285.6$ of the frequency-domain solver with the analytic impedance, while f_0 changes by 0.10%. The largest run is consistent with convergence towards that value but does not demonstrate it.

6 Results

6.1 The resonance and the end correction

In the time domain the resonator selects its frequency: after the pulse has passed, the cavity rings at f_0 with no frequency imposed anywhere (Fig. 5a). In the frequency domain the response is narrow: on the 1 mm grid the −3 dB width is 0.73 Hz. A first sweep at 1 Hz steps read the peak at 209.95 Hz and 6.62 kPa; at 0.005 Hz steps it is 209.84 Hz and 7.25 kPa, a 9% error in amplitude because the step was wider than the peak.

Inverting Eq. (1) gives the end correction implied by each frequency. From the raw 1 mm time-domain value it is $1.29a$, 15% below the $1.51a$ of the reference; from the extrapolated value it is $1.56a$, about 3% above it. The apparent deficit of the meshed exterior was the wall offset.

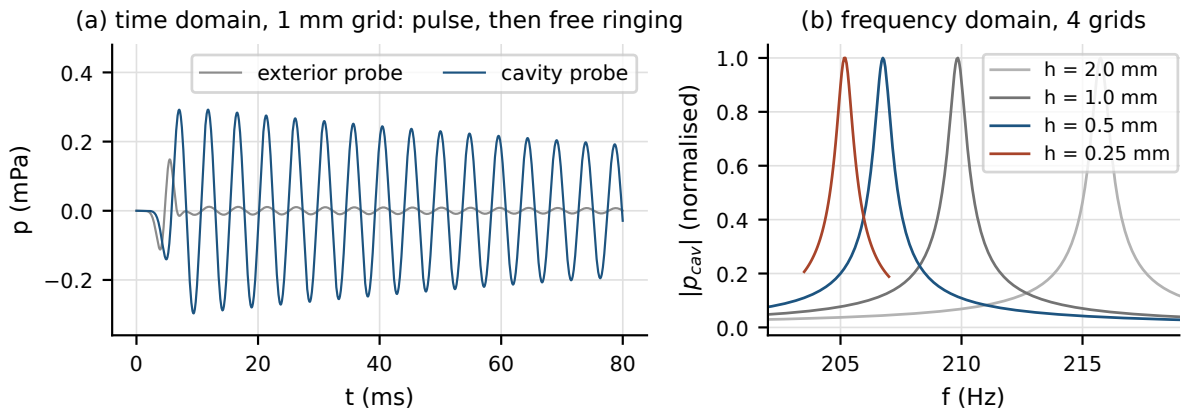


Figure 5: (a) Time-domain solver, 1 mm grid: the pulse reaches the exterior probe, then the cavity keeps oscillating at its own frequency (`data/open_resonator.npz`). (b) Frequency-domain response of the cavity on successive grids (`data/fdm_sweep_h*.npz`); the peak moves down as the walls move in.

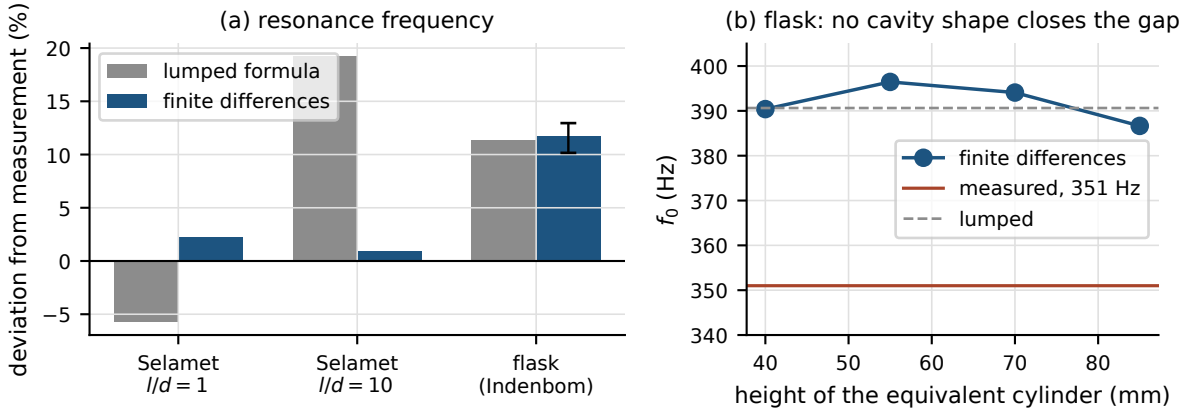


Figure 6: Comparison with published measurements (`data/literature_validation.npz`; Selamet values from `notebooks/harmonic_study.ipynb`, section 5). (a) Deviation of the lumped formula and of the finite-difference solver; the bar on the flask spans the cavity aspect ratios tried. (b) Flask: finite-difference f_0 against the height of the equivalent cylindrical cavity.

With the impedance condition, the frequency-domain study gives a shift in ΔL of $0.823a$ between the Dirichlet and the radiating mouth, against $8a/(3\pi) = 0.849a$ in Eq. (4). Since that reactance is imposed, this is a consistency check of the chain, not an independent computation.

6.2 Comparison with published measurements

Selamet et al. measured two concentric resonators with the same 4500 cm^3 cavity in two shapes ($l/d = 1$ and 10) and a neck of diameter 40.44 mm and length 85 mm [19]. They are side branches on a duct, so the neck sees a pressure node at resonance; they were computed with the Dirichlet mouth. The lumped formula knows only the volume and returns 85.8 Hz for both, -5.7% and $+19\%$ from the measurements. The two-dimensional field recovers both to 2.2% and 0.9% (Table 4). These values are on the 1 mm grid, not extrapolated, and the deviations are of the size of the GCI.

Indenbom and Pogossian measured an empty 100 mL Erlenmeyer flask at 351 Hz [6], where the lumped formula gives 391 Hz; my lumped model reproduces 390.6 Hz. The flask is conical and my cavity is a cylinder, so I replaced it by a cylinder of equal volume and swept its aspect ratio. The solver stays between 387 and 396 Hz for every shape (Fig. 6b), 10–13% above the measurement. Resolving the field near the neck does not close the gap. In terms of Eq. (1), the

Table 4: Resonance frequencies (Hz) against published measurements. Finite differences on the 1 mm grid, no fitted parameter.

case	measured	lumped, Eq. (1)	finite differences	deviation
Selamet et al., $l/d = 1$	91	85.8	93.0	+2.2%
Selamet et al., $l/d = 10$	72	85.8	72.7	+0.9%
Indenbom & Pogossian, flask	351	390.6	387–396	+10 to +13%

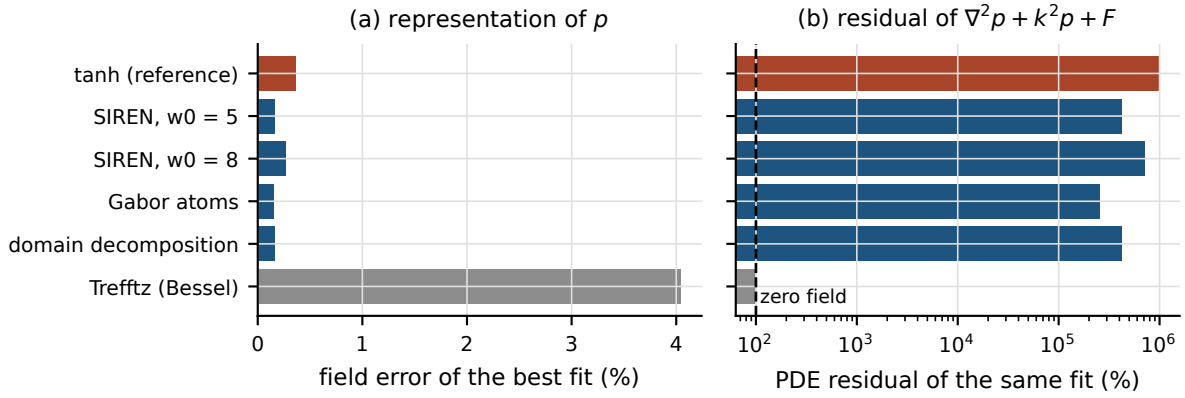


Figure 7: Least-squares projection of the verified field onto 96 basis functions of each family (`data/bench_bases.npz`). (a) Relative field error. (b) PDE residual of the same projection, normalised so that the zero field scores 100% (dashed line).

measurement corresponds to an effective neck of about 54 mm instead of 43 mm, which would be consistent with the neck flaring into the conical shoulder, a geometry this solver cannot represent. I report this as an unexplained miss.

For damping, the same paper gives a half-width that brackets the measured Q between 18 and 30 (the paper does not state whether the width is taken at half amplitude or half power). An estimate combining viscothermal losses in the neck and radiation gives about 40: the model is optimistic, in the same direction as Moloney's comparison [9]. Neither solver contains viscothermal losses.

6.3 Neural bases: a measured negative result

All neural bases represent the field well: the field error of the projection is between 0.15% and 0.36% (Fig. 7a). None represents its Laplacian: the residual of the same projection is between 2,536 and 9,979 times that of the zero field (Fig. 7b). The conditioning of the residual operator varies by eleven orders of magnitude between families (from 5×10^2 to 7×10^{13}), with sine activations far better conditioned than tanh, yet the residual barely improves. The cause is the structure of the solution: inside the cavity $\nabla^2 p$ and $-k^2 p$ are both large and nearly cancel, so a 0.15% error on p becomes an error on $\nabla^2 p$ larger than the source itself. The Trefftz basis is the counter-example: each function satisfies $\nabla^2 \phi + k^2 \phi = 0$ exactly, so its residual equals that of the zero field (100%), but it cannot represent the source and fits the field only to 4.0%.

7 Discussion and limitations

What the results support. Two solvers with different discretisations and different radiation models agree on the natural frequency of the reference resonator to 0.6%, inside their grid uncertainties of about 2%, and the frequency-domain scheme reproduces two published resonators to within that uncertainty. The verification sequence found an error that the cross-check alone

would have hidden: both solvers carry the same half-cell wall offset, so their raw values agree better than either agrees with its own converged value.

What they do not support.

- *Damping.* Both solvers are lossless. The viscothermal boundary layer in the neck (about 0.14 mm at this frequency) is the dominant loss and is only estimated analytically. The Q values above are radiation-only and are not predictions of a real resonator.
- *Grid convergence.* Observed orders near 1 come from the wall offset and the staircase walls, so the GCI is a first-order estimate. The time-domain extrapolation rests on three grids.
- *Geometry.* Cavities are cylinders. The flask miss is consistent with its conical shoulder, but this has not been shown.
- *Outer boundary.* The Cartesian PML has a 4% reflection floor on the curved boundary, and the reflection was measured with a 4 kHz-centred pulse, whose wavelength is much shorter than at the resonance. The radiation Q of the time-domain solver is not converged in exterior size.
- *Comparison data.* All measurements are other authors'. Two of the three cases are side-branch resonators computed with a Dirichlet mouth on a single grid.
- *Neural bases.* The test measures projections onto fixed random bases, not trained networks, and the target is the 1 mm finite-difference field, which itself satisfies the continuous equation only to discretisation accuracy.

8 Conclusion and future work

A finite-difference model of an open Helmholtz resonator predicts its natural frequency with a stated uncertainty of about 2%, provided that the half-cell wall offset revealed by the manufactured solution is extrapolated away. The end correction computed by the meshed exterior, $1.56a$, is within about 3% of the semi-empirical value, and two published resonators are reproduced to 0.9–2.2%. One published resonator is missed by about 11%, and damping is outside the scope of these lossless models. The next steps are a cylindrical PML to remove the 4% floor, a body-fitted or cut-cell wall treatment to recover second-order convergence, a viscothermal boundary-layer model in the neck, and a ring-down measurement of a real resonator, for which the protocol and analysis script are in the repository.

Reproducibility statement

All code, data and figures are in the repository. The figures and the numbers quoted in the text are produced by `python paper/make_figures.py` from versioned files in `data/`, and the report is built by `paper/build_paper.sh`. Main commands and times on the machine used (Intel Core i7-6700HQ, 4 cores, 32 GB RAM, Windows 10; Python 3.11.9, NumPy 2.2.6, SciPy 1.15.2, Matplotlib 3.10.6, PyTorch 2.8.0 CPU; one core per run):

- `verification/mms_transient.py`: transient MMS, 15 s.
- `src/transient_solver.py` with `OR_TMS=250` and `OR_H` set to 2, 1 and 0.5: the three time-domain runs, about 1, 5 and 40 min (scaled from timed short runs); then `verification/grid_convergence`
- `studies/frequency_sweep.py` with `SW_DF=0.05`: over 195–220 Hz with `SW_H` set to 2, 1 and 0.5 (under 1, 1 and 6 min), and over 203.5–207 Hz with `SW_H=0.25` (71 solves, 9 min on a loaded machine); then `verification/control_grid.py`.

- `studies/radiation_boundaries.py`: part A (reflection) a few minutes, parts B and C several hours.
- `experiments/literature_validation.py`: about 4 min.
- `studies/neural_basis_benchmark.py`: under 1 min.
- `notebooks/harmonic_study.ipynb`: Dirichlet-mouth study, MMS, Selamet cases.

References

[1] J.-P. Bérenger. A perfectly matched layer for the absorption of electromagnetic waves. *Journal of Computational Physics*, 114(2):185–200, 1994. doi: 10.1006/jcph.1994.1159.

[2] I. B. Celik, U. Ghia, P. J. Roache, C. J. Freitas, H. Coleman, and P. E. Raad. Procedure for estimation and reporting of uncertainty due to discretization in CFD applications. *Journal of Fluids Engineering*, 130(7):078001, 2008. doi: 10.1115/1.2960953.

[3] F. Collino and P. Monk. The perfectly matched layer in curvilinear coordinates. *SIAM Journal on Scientific Computing*, 19(6):2061–2090, 1998. doi: 10.1137/S1064827596301406.

[4] M. J. Grote and I. Sim. Efficient PML for the wave equation. arXiv:1001.0319, 2010.

[5] R. Hiptmair, A. Moiola, and I. Perugia. A survey of Trefftz methods for the Helmholtz equation. In *Building Bridges: Connections and Challenges in Modern Approaches to Numerical Partial Differential Equations*, Lecture Notes in Computational Science and Engineering, pages 237–279. Springer, 2016. doi: 10.1007/978-3-319-41640-3_8.

[6] M. V. Indenbom and S. P. Pogossian. Characteristics of partially filled Helmholtz resonators. *Acta Acustica*, 7:51, 2023. doi: 10.1051/aacus/2023039. arXiv:2212.08858.

[7] U. Ingard. On the theory and design of acoustic resonators. *The Journal of the Acoustical Society of America*, 25(6):1037–1061, 1953. doi: 10.1121/1.1907235.

[8] Lord Rayleigh. *The Theory of Sound*, volume 2. Macmillan, London, 2nd edition, 1896.

[9] M. J. Moloney. Quality factors and conductances in Helmholtz resonators. *American Journal of Physics*, 72(8):1035–1039, 2004. doi: 10.1119/1.1677375.

[10] P. M. Morse and K. U. Ingard. *Theoretical Acoustics*. McGraw-Hill, New York, 1968.

[11] B. Moseley, A. Markham, and T. Nissen-Meyer. Finite basis physics-informed neural networks (FBPINNs): a scalable domain decomposition approach for solving differential equations. *Advances in Computational Mathematics*, 49(4), 2023. doi: 10.1007/s10444-023-10065-9.

[12] G. Mur. Absorbing boundary conditions for the finite-difference approximation of the time-domain electromagnetic-field equations. *IEEE Transactions on Electromagnetic Compatibility*, EMC-23(4):377–382, 1981. doi: 10.1109/TEMC.1981.303970.

[13] W. L. Oberkampf and C. J. Roy. *Verification and Validation in Scientific Computing*. Cambridge University Press, 2010. doi: 10.1017/CBO9780511760396.

[14] N. Rahaman, A. Baratin, D. Arpit, F. Draxler, M. Lin, F. A. Hamprecht, Y. Bengio, and A. Courville. On the spectral bias of neural networks. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019. arXiv:1806.08734.

[15] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.2018.10.045.

[16] P. J. Roache. Perspective: A method for uniform reporting of grid refinement studies. *Journal of Fluids Engineering*, 116(3):405–413, 1994. doi: 10.1115/1.2910291.

[17] P. J. Roache. *Verification and Validation in Computational Science and Engineering*. Hermosa Publishers, Albuquerque, NM, 1998.

[18] P. J. Roache. Code verification by the method of manufactured solutions. *Journal of Fluids Engineering*, 124(1):4–10, 2002. doi: 10.1115/1.1436090.

[19] A. Selamet, P. M. Radavich, N. S. Dickey, and J. M. Novak. Circular concentric Helmholtz resonators. *The Journal of the Acoustical Society of America*, 101(1):41–51, 1997. doi: 10.1121/1.417986.

[20] V. Sitzmann, J. N. P. Martel, A. W. Bergman, D. B. Lindell, and G. Wetzstein. Implicit neural representations with periodic activation functions. arXiv:2006.09661, 2020.

[21] H. von Helmholtz. Theorie der Luftschwingungen in Röhren mit offenen Enden. *Journal für die reine und angewandte Mathematik*, 57:1–72, 1860. doi: 10.1515/crll.1860.57.1.

[22] S. Wang, H. Wang, and P. Perdikaris. On the eigenvector bias of Fourier feature networks: From regression to solving multi-scale PDEs with physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering*, 384:113938, 2021. doi: 10.1016/j.cma.2021.113938.